Care Transition Efficiency & Placement Outcome Analytics

Abstract

This paper evaluates the efficiency of the Unaccompanied Children (UAC) care transition pipeline modeled as CBP Custody → HHS Care → Sponsor Placement. Exploratory Data Analysis (EDA), derived process metrics, backlog monitoring, and interactive visualization are used to identify bottlenecks and assess placement outcomes. A Streamlit dashboard was developed to support operational monitoring and decision-making.

I. Introduction

The UAC program requires efficient transitions between CBP custody, HHS care, and sponsor placement. Monitoring operational efficiency helps reduce delays and improve reunification outcomes.

II. Problem Statement

Traditional reporting emphasizes custody counts while providing limited visibility into transfer efficiency, discharge effectiveness, and backlog accumulation.

III. Methodology

Data preprocessing included date conversion, numeric cleaning, missing-value handling, and feature engineering. Key metrics included Transfer Efficiency, Discharge Efficiency, Pipeline Throughput, Placement Ratio, Daily Backlog, and Cumulative Backlog. Temporal analysis, bottleneck detection, and outcome stability analyses were performed.

IV. Exploratory Data Analysis

EDA examined pipeline flow, efficiency trends, backlog behavior, monthly placement outcomes, and sudden changes in reunification performance. Interactive Plotly visualizations and Streamlit components supported exploration.

V. Results and Discussion

The analytical framework enables identification of operational bottlenecks, periods of increasing backlog, and fluctuations in placement performance. Transfer and discharge metrics provide actionable operational KPIs.

VI. Recommendations

Increase sponsor verification capacity during demand surges, deploy automated backlog alerts, integrate predictive analytics, and collect facility-level processing metrics.

VII. Conclusion

Modeling the UAC process as a transition pipeline provides deeper operational insight than aggregate counts alone. The proposed dashboard supports data-driven monitoring and continuous process improvement.

References

[1] U.S. Department of Health and Human Services (HHS).
[2] U.S. Customs and Border Protection (CBP).
[3] Streamlit Documentation.
[4] Pandas Documentation.
[5] Plotly Documentation.